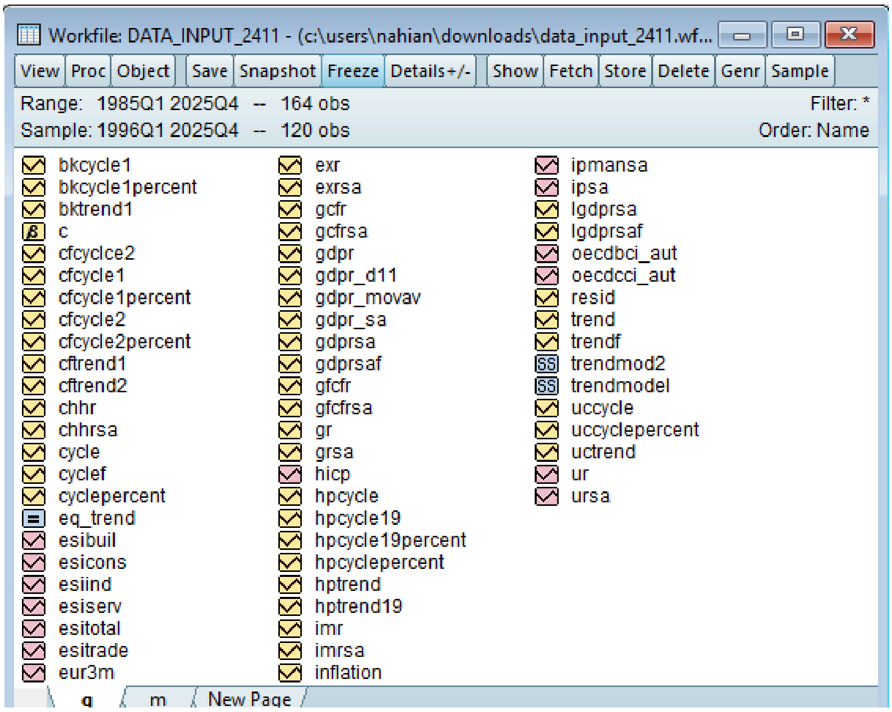
Assessment of Leading Indicators for Austrian GDP and Private Consumption

The assessment requires assessing the nowcasting or forecasting performance of different indicators for Austrian GDP (try levels, qoq growth rates, yoy growth rates):
- OECD Consumer Confidence Indicator (OECDCCI) (for private consumption)
- OECD Business Confidence Indicator (OECDCCI) (for GDP)
- European Commission Economic Sentiment Indicator (for GDP)
- European Commission Economic Sentiment Indicator Consumers (for private consumption).

The following is the workfile that will be used -



We work with samples from 1996Q1 to 2025Q4. The variables that will be used -
GDP seasonally adjusted = **gdprsa**
Private Consumption = **chhrsa**
OECD Consumer Confidence Indicator for Austria = **oecdcci_aut**
OECD Business Confidence Indicator for Austria = **oecdbci_aut**
European Commission Economic Sentiment Indicator = **esitotal**
European Commission Economic Sentiment Indicator Consumers = **esicons**

Why Do We Need Leading Indicators?

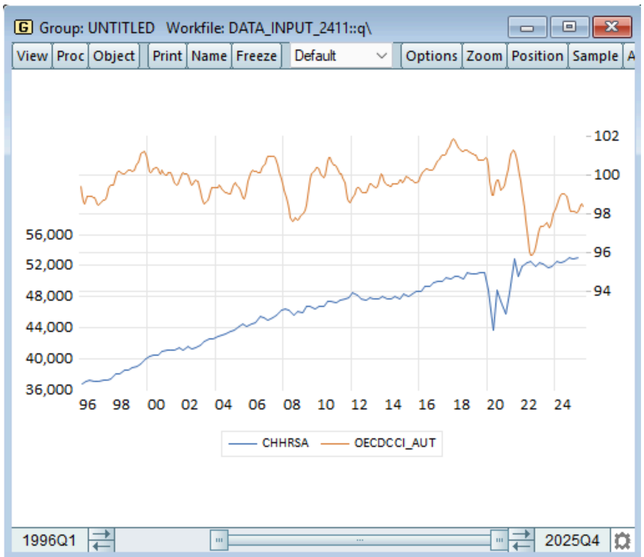
Official macroeconomic statistics like GDP and private consumption are released with long publication lags—typically 45 to 60 days after the end of the quarter. In contrast, sentiment indicators (OECD and EC) are available monthly, providing early soft information on turning points in economic activity. Policymakers, central banks, and financial institutions rely on these indicators to anticipate recessions, gauge the strength of recoveries, and adjust monetary or fiscal decisions before official data arrives. Leading indicators therefore fill the real-time information gap in macroeconomic monitoring.

1. OECD Consumer Confidence Indicator (OECDCCI) for Private Consumption

The analysis of the OECD Consumer Confidence Indicator as a predictor of Austrian private consumption reveals dramatically different results across specifications, demonstrating that proper data transformation is critical for valid econometric inference.

1.1 Levels

Graph



Cross-correlation

Group: UNTITLED Workfile: DATA_INPUT_2411::q\

Cross Correlogram of CHHRSA and OECDCCI_AUT

Date: 11/28/25 Time: 18:18
Sample: 1996Q1 2025Q4
Included observations: 118
Correlations are asymptotically consistent approximations

CHHRSA,OECDCCI_AUT...	CHHRSA,OECDCCI_AUT...	i	lag	lead
		0	-0.0867	-0.0867
		1	-0.0720	-0.1183
		2	-0.0553	-0.1439
		3	-0.0404	-0.1600
		4	-0.0443	-0.1697

Granger Casualty

Group: UNTITLED Workfile: DATA_INPUT_2411::q\

Pairwise Granger Causality Tests
Date: 11/28/25 Time: 18:19
Sample: 1996Q1 2025Q4
Lags: 4

Null Hypothesis:	Obs	F-Statistic	Prob.
OECDCCI_AUT does not Granger Cause CHHRSA	118	0.34279	0.8485
CHHRSA does not Granger Cause OECDCCI_AUT		6.26172	0.0001

Equation

Equation: UNTITLED Workfile: DATA_INPUT_2411::q\

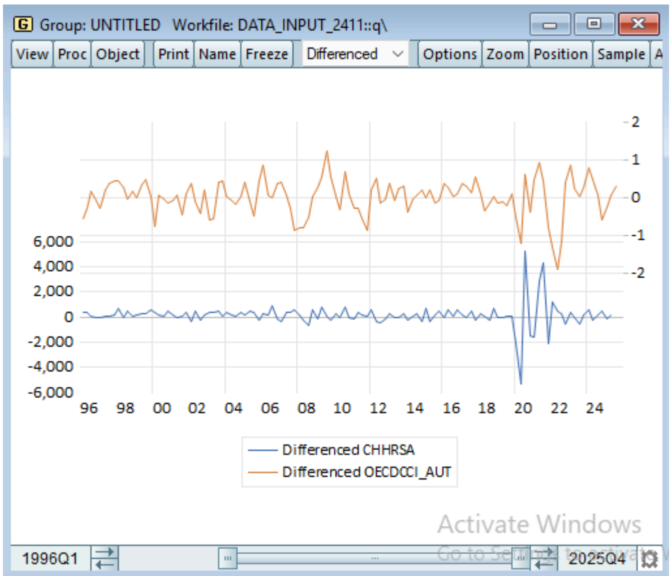
Dependent Variable: CHHRSA
Method: Least Squares
Date: 11/28/25 Time: 18:19
Sample (adjusted): 1996Q1 2025Q2
Included observations: 118 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	82759.01	39281.32	2.106829	0.0373
OECDCCI_AUT	-369.8908	394.4544	-0.937728	0.3503

R-squared	0.007523	Mean dependent var	45926.01
Adjusted R-squared	-0.001032	S.D. dependent var	4631.537
S.E. of regression	4633.927	Akaike info criterion	19.73700
Sum squared resid	2.49E+09	Schwarz criterion	19.78396
Log likelihood	-1162.483	Hannan-Quinn criter.	19.75607
F-statistic	0.879333	Durbin-Watson stat	0.049813
Prob(F-statistic)	0.350333		

The levels specification completely fails, producing negative cross-correlations (-0.09 to -0.17), no Granger causality (p-value = 0.848), an insignificant negative coefficient (-369.89, p-value = 0.350), and negligible explanatory power (R^2 = 0.75%) with severe autocorrelation (Durbin-Watson = 0.05), confirming this is a spurious regression caused by mixing a trending consumption series with a mean-reverting confidence index. This specification violates basic econometric principles by attempting to relate consumption levels—determined by long-run wealth and demographics—to a cyclical sentiment indicator, making it entirely unsuitable for forecasting.

1.2 QoQ Growth Rates



Equation: UNTITLED Workfile: DATA_INPUT_2411::q\

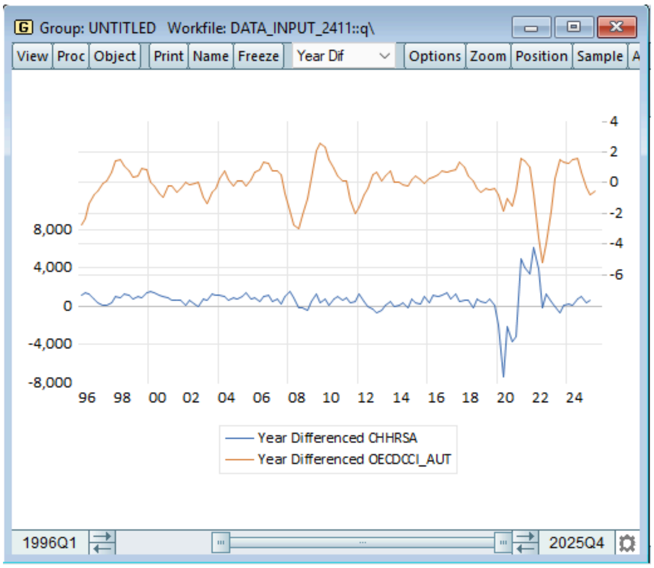
Dependent Variable: @PC(CHHRSA)
Method: Least Squares
Date: 11/28/25 Time: 18:38
Sample (adjusted): 1996Q1 2025Q2
Included observations: 118 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.351009	0.181557	1.933325	0.0556
@PC(OECDCCI_AUT)	1.136570	0.378010	3.006720	0.0032

R-squared	0.072300	Mean dependent var	0.338338
Adjusted R-squared	0.064302	S.D. dependent var	2.038306
S.E. of regression	1.971683	Akaike info criterion	4.212456
Sum squared resid	450.9540	Schwarz criterion	4.259417
Log likelihood	-246.5349	Hannan-Quinn criter.	4.231523
F-statistic	9.040366	Durbin-Watson stat	2.438193
Prob(F-statistic)	0.003238		

The quarter-on-quarter growth rate specification transforms the indicator into a valid and effective predictor, producing a positive, highly significant coefficient of 1.137 (t-statistic = 3.007, p-value = 0.003) that aligns with economic theory. The model explains 7.2% of consumption growth variation, which is appropriate given that consumption depends on multiple factors beyond sentiment, and passes diagnostic tests with an acceptable Durbin-Watson statistic of 2.438. This specification correctly captures how confidence affects short-run consumption decisions through precautionary savings adjustments and timing of discretionary purchases, making it suitable for nowcasting applications.

1.3 YoY Growth Rates



Equation: UNTITLED Workfile: DATA_INPUT_2411::q\

Dependent Variable: @PCY(CHHRSA)
Method: Least Squares
Date: 11/28/25 Time: 18:47
Sample (adjusted): 1996Q1 2025Q2
Included observations: 118 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1.375293	0.264057	5.208319	0.0000
@PCY(OECDCCI_AUT)	0.284072	0.210435	1.349924	0.1797

R-squared	0.015466	Mean dependent var	1.355714
Adjusted R-squared	0.006979	S.D. dependent var	2.874110
S.E. of regression	2.864063	Akaike info criterion	4.959164
Sum squared resid	951.5316	Schwarz criterion	5.006124
Log likelihood	-290.5907	Hannan-Quinn criter.	4.978231
F-statistic	1.822295	Durbin-Watson stat	0.835360
Prob(F-statistic)	0.179669		

The year-on-year growth rate specification, while properly achieving stationarity, shows surprisingly weak performance with an insignificant coefficient (p-value = 0.180) and explains only 1.5% of variation despite passing diagnostic tests (Durbin-Watson = 1.822). The poor performance likely stems from annual differencing smoothing away high-frequency variation where confidence effects are strongest, as sentiment shifts drive quarterly spending decisions but dissipate when aggregated to annual horizons. The YoY transformation may also suffer from excessive data overlap, reducing effective information content and allowing confounding factors to obscure the confidence-consumption relationship.

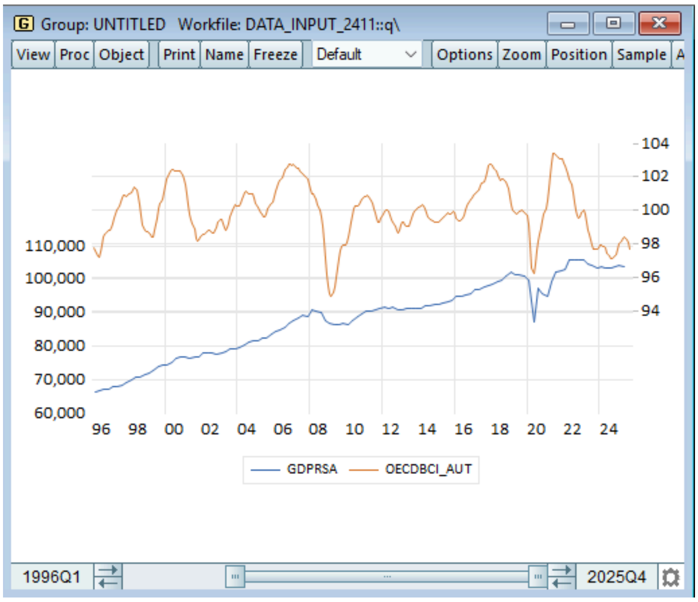
For policy and forecasting purposes, the quarter-on-quarter specification should be strongly preferred as it isolates the cyclical component where confidence operates, provides statistically significant results, and aligns with theoretical mechanisms linking expectations to near-term behavior. The indicator's primary value lies in its timeliness (monthly availability) rather than standalone explanatory power, making it particularly useful for filling the 45-60 day information gap before official consumption data is published and for detecting turning points during economic transitions. Overall, the OECD Consumer Confidence Indicator demonstrates moderate but genuine forecasting utility when properly specified in quarterly growth rates, though it should be combined with other predictors for robust consumption forecasting.

2. OECD Business Confidence Indicator (OECDBCI) for GDP

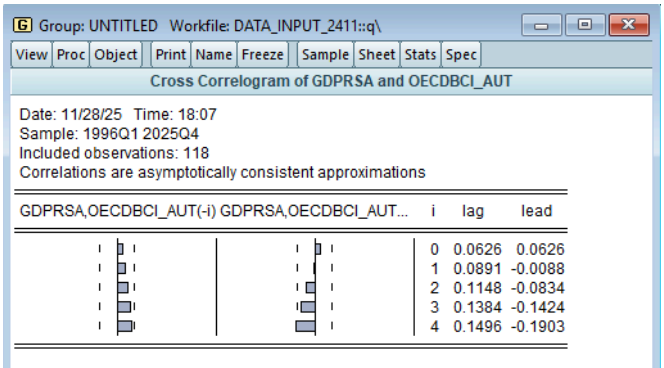
The analysis of the OECD Business Confidence Indicator as a predictor of Austrian GDP demonstrates a stark transformation from complete econometric failure in levels to robust and significant forecasting power in growth rates, with notably stronger performance at annual horizons compared to consumer confidence, highlighting both the critical importance of proper stationarity transformations and the distinct channels through which business sentiment influences aggregate economic activity.

<u>2.1 Levels</u>

Graph



Cross Correlation



Granger Casualty

Group: UNTITLED Workfile: DATA_INPUT_2411::q\

View | Proc | Object | Print | Name | Freeze | Sample | Sheet | Stats | Spec

Pairwise Granger Causality Tests
Date: 11/28/25 Time: 18:07
Sample: 1996Q1 2025Q4
Lags: 4

Null Hypothesis:	Obs	F-Statistic	Prob.
OECDBCI_AUT does not Granger Cause GDPRSA	118	3.99013	0.0046
GDPRSA does not Granger Cause OECDBCI_AUT		2.61172	0.0393

Equation

Equation: UNTITLED Workfile: DATA_INPUT_2411::q\

View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

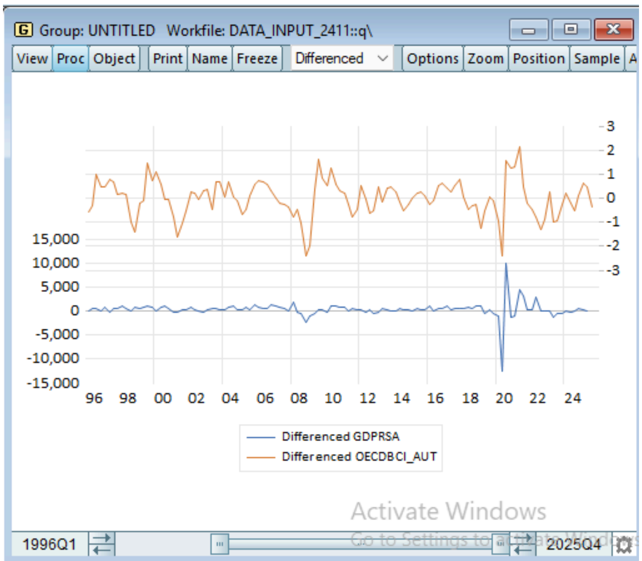
Dependent Variable: GDPRSA
Method: Least Squares
Date: 11/28/25 Time: 18:09
Sample (adjusted): 1996Q1 2025Q2
Included observations: 118 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	45580.58	63028.62	0.723173	0.4710
OECDBCI_AUT	426.2971	630.6271	0.675989	0.5004

R-squared	0.003924	Mean dependent var	88181.62
Adjusted R-squared	-0.004663	S.D. dependent var	11103.20
S.E. of regression	11129.06	Akaike info criterion	21.48931
Sum squared resid	1.44E+10	Schwarz criterion	21.53627
Log likelihood	-1265.869	Hannan-Quinn criter.	21.50838
F-statistic	0.456961	Durbin-Watson stat	0.020880
Prob(F-statistic)	0.500393		

The levels specification fails entirely, producing an insignificant coefficient of 426.30 (p-value = 0.5004), near-zero explanatory power (R^2 = 0.4%, adjusted R^2 = -0.4%), severe autocorrelation (Durbin-Watson = 0.021), and inconsistent cross-correlations that turn negative at longer lags, confirming this is a spurious regression mixing a trending GDP series with a mean-reverting confidence index. This specification violates fundamental econometric principles by attempting to relate GDP levels—determined by long-run supply factors like capital, labor, and technology—to a cyclical sentiment indicator that has no theoretical basis for explaining the economy's absolute productive capacity.

2.2. QoQ Growth Rates



Equation: UNTITLED Workfile: DATA_INPUT_2411::q\

View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

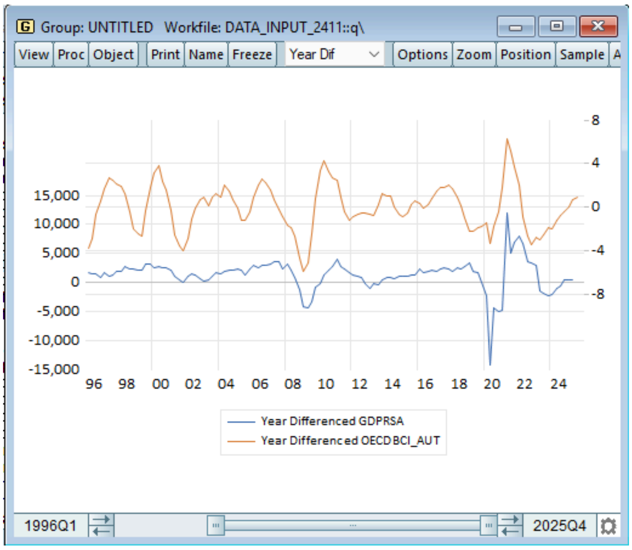
Dependent Variable: @PC(GDPRSA)
Method: Least Squares
Date: 11/28/25 Time: 18:40
Sample (adjusted): 1996Q1 2025Q2
Included observations: 118 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.393106	0.147214	2.670304	0.0087
@PC(OECDBCI_AUT)	1.101036	0.193272	5.696813	0.0000

R-squared	0.218611	Mean dependent var	0.398294
Adjusted R-squared	0.211875	S.D. dependent var	1.801293
S.E. of regression	1.599123	Akaike info criterion	3.793592
Sum squared resid	296.6346	Schwarz criterion	3.840553
Log likelihood	-221.8219	Hannan-Quinn criter.	3.812659
F-statistic	32.45368	Durbin-Watson stat	2.597236
Prob(F-statistic)	0.000000		

The quarter-on-quarter growth rate specification transforms the indicator into a statistically valid predictor, producing a highly significant positive coefficient of 1.110 (t-statistic = 5.696, p-value = 0.0000) with moderate explanatory power (R^2 = 21.9%, adjusted R^2 = 21.2%) and acceptable diagnostics (Durbin-Watson = 2.597). The time series plots show clear synchronous movements, particularly during the 2020 COVID-19 crisis, and the significant relationship confirms that when entrepreneurs are optimistic about future demand, they increase production, investment, and hiring, thereby driving current quarter GDP growth. However, the modest R-squared indicates that business confidence captures only part of the GDP growth story, as output is also driven by external demand shocks, fiscal policy, monetary conditions, and productivity developments.

2.3 YoY Growth Rates



Equation: UNTITLED Workfile: DATA_INPUT_2411::q\

View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

Dependent Variable: @PCY(GDPRSA)
Method: Least Squares
Date: 11/28/25 Time: 18:49
Sample (adjusted): 1996Q1 2025Q2
Included observations: 118 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1.648235	0.235090	7.011092	0.0000
@PCY(OECDBCI_AUT)	0.671790	0.109383	6.141618	0.0000

R-squared	0.245379	Mean dependent var	1.617732
Adjusted R-squared	0.238873	S.D. dependent var	2.926503
S.E. of regression	2.553157	Akaike info criterion	4.729342
Sum squared resid	756.1587	Schwarz criterion	4.776303
Log likelihood	-277.0312	Hannan-Quinn criter.	4.748409
F-statistic	37.71947	Durbin-Watson stat	0.809194
Prob(F-statistic)	0.000000		

The year-on-year growth rate specification delivers surprisingly strong results, substantially outperforming both the YoY consumer confidence model and even slightly exceeding the QoQ business confidence model, with a highly significant coefficient of 1.648 (t-statistic = 7.011, p-value = 0.0000) and robust explanatory power (R^2 = 24.5%, adjusted R^2 = 23.9%). This superior performance at annual horizons is theoretically sensible because business investment decisions operate with longer planning horizons than consumer spending—capital projects require extended implementation periods and entrepreneurs form expectations about sustained demand conditions over multiple quarters rather than just the immediate future. The Durbin-Watson statistic of 0.809 suggests some remaining autocorrelation, indicating that additional lags or complementary predictors might improve the model, but does not invalidate the strong significant relationship given the highly significant t-statistics.

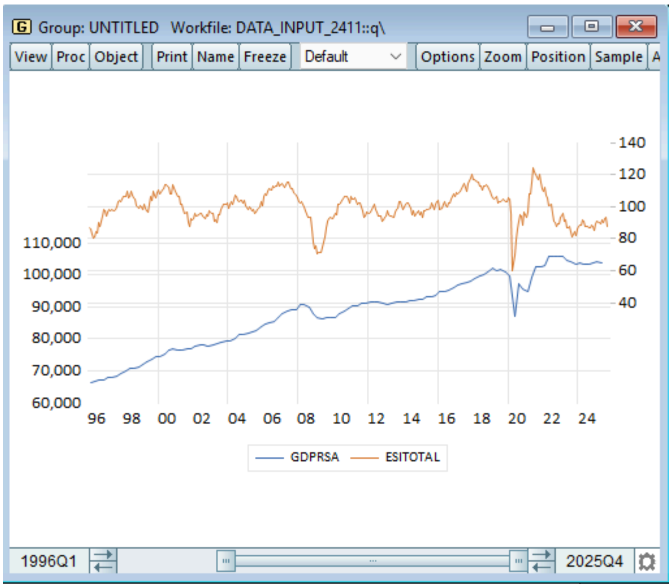
The contrast between consumer and business confidence is revealing: consumer confidence showed strong QoQ performance (R^2 = 7.2%) but collapsed at annual horizons (R^2 = 1.5%), while business confidence maintains robust performance at both frequencies and actually strengthens at annual horizons, reflecting that business sentiment drives longer-lasting decisions about capacity, employment, and investment that shape GDP over multiple quarters. For practical forecasting applications, the OECD Business Confidence Indicator demonstrates genuine utility across both quarterly and annual horizons, making it superior to consumer confidence for GDP nowcasting and establishing it as a valuable leading indicator that should be incorporated into multivariate forecasting models alongside hard indicators like industrial orders, credit growth, and external demand. The indicator's primary strength lies in capturing entrepreneurial expectations about demand conditions that presage behavioral changes in production and investment, though it should be combined with other predictors capturing supply constraints, financial conditions, and fiscal/monetary policy for comprehensive GDP forecasts that explain the remaining 75-80% of growth variation driven by factors beyond business sentiment.

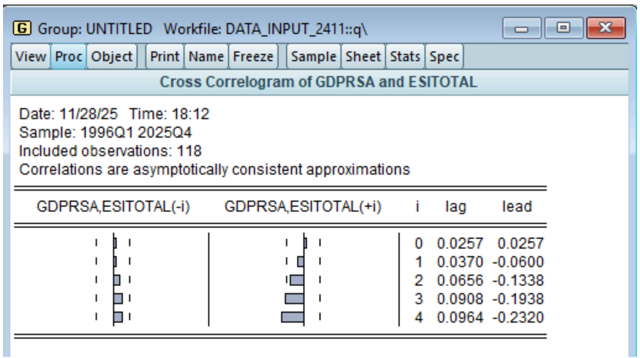
3. European Commission Economic Sentiment Indicator for GDP

The analysis of the European Commission Economic Sentiment Indicator as a predictor of Austrian GDP reveals a pattern strikingly similar to the OECD Business Confidence Indicator, with complete failure in levels but exceptional performance in growth rates, particularly demonstrating the strongest results of all indicators examined when properly transformed to capture cyclical dynamics.

3.1 Levels

Graph **Cross-correlation**



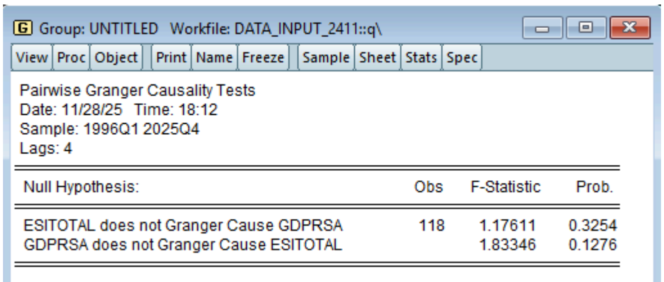


Cross Correlogram of GDPRSA and ESITOTAL

Date: 11/28/25 Time: 18:12
Sample: 1996Q1 2025Q4
Included observations: 118
Correlations are asymptotically consistent approximations

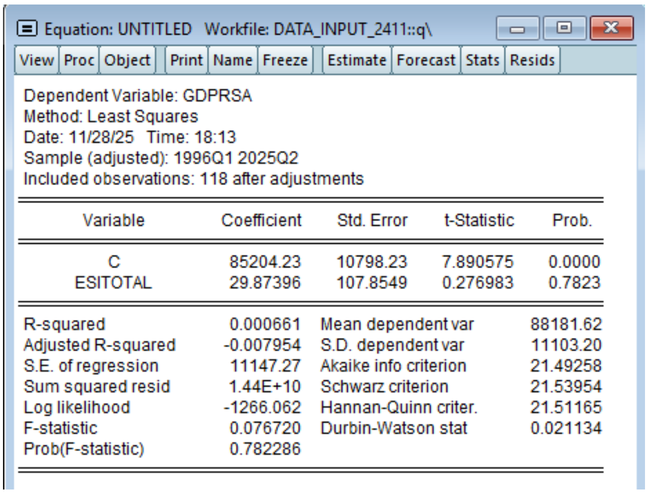
GDPRSA,ESITOTAL(-i)	GDPRSA,ESITOTAL(+i)	i	lag	lead
		0	0.0257	0.0257
		1	0.0370	-0.0600
		2	0.0656	-0.1338
		3	0.0908	-0.1938
		4	0.0964	-0.2320

Granger casualty **Equation**



Pairwise Granger Causality Tests
Date: 11/28/25 Time: 18:12
Sample: 1996Q1 2025Q4
Lags: 4

Null Hypothesis:	Obs	F-Statistic	Prob.
ESITOTAL does not Granger Cause GDPRSA	118	1.17611	0.3254
GDPRSA does not Granger Cause ESITOTAL		1.83346	0.1276



Dependent Variable: GDPRSA
Method: Least Squares
Date: 11/28/25 Time: 18:13
Sample (adjusted): 1996Q1 2025Q2
Included observations: 118 after adjustments

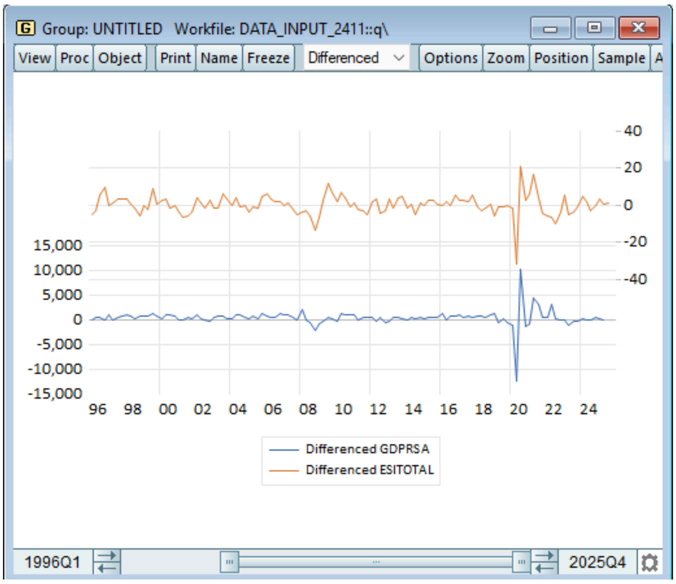
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	85204.23	10798.23	7.890575	0.0000
ESITOTAL	29.87396	107.8549	0.276983	0.7823

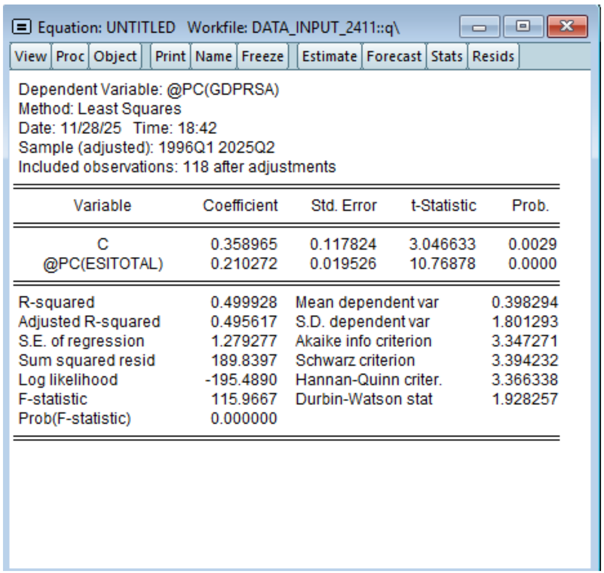
R-squared	0.000661	Mean dependent var	88181.62
Adjusted R-squared	-0.007954	S.D. dependent var	11103.20
S.E. of regression	11147.27	Akaike info criterion	21.49258
Sum squared resid	1.44E+10	Schwarz criterion	21.53954
Log likelihood	-1266.062	Hannan-Quinn criter.	21.51165
F-statistic	0.076720	Durbin-Watson stat	0.021134
Prob(F-statistic)	0.782286		

The levels specification is econometrically invalid, producing weak and inconsistent cross-correlations (0.0257 at lag 0, then turning negative at leads with -0.0600 at lag 1 and -0.2320 at lag 4), marginally significant but theoretically problematic Granger causality in both directions (ESI to GDP: F = 1.1761, p = 0.3254; GDP to ESI: F = 1.833, p = 0.1276), and a regression with a positive but insignificant coefficient of 29.87 (t-statistic = 0.276, p =

0.7823) that explains essentially nothing (R^2 = 0.07%, adjusted R^2 = -0.8%). The Durbin-Watson statistic of 0.021 reveals severe positive autocorrelation identical to previous levels specifications, confirming this is another spurious regression mixing a trending GDP series with a mean-reverting sentiment index around 100, violating the fundamental principle that GDP levels are determined by long-run supply capacity rather than cyclical sentiment fluctuations.

3.2 QoQ Growth Rates





Equation: UNTITLED Workfile: DATA_INPUT_2411::q\

View Proc Object Print Name Freeze Estimate Forecast Stats Resids

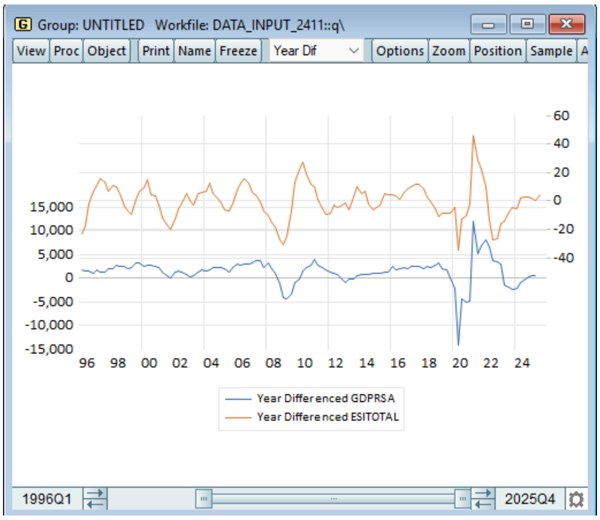
Dependent Variable: @PC(GDPRSA)
Method: Least Squares
Date: 11/28/25 Time: 18:42
Sample (adjusted): 1996Q1 2025Q2
Included observations: 118 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.358965	0.117824	3.046633	0.0029
@PC(ESITOTAL)	0.210272	0.019526	10.76878	0.0000

R-squared	0.499928	Mean dependent var	0.398294
Adjusted R-squared	0.495617	S.D. dependent var	1.801293
S.E. of regression	1.279277	Akaike info criterion	3.347271
Sum squared resid	189.8397	Schwarz criterion	3.394232
Log likelihood	-195.4890	Hannan-Quinn criter.	3.366338
F-statistic	115.9667	Durbin-Watson stat	1.928257
Prob(F-statistic)	0.000000		

The quarter-on-quarter growth rate specification transforms the ESI into an exceptionally powerful predictor, producing a highly significant positive coefficient of 0.210 (t-statistic = 10.769, p = 0.0000) with the strongest explanatory power of any specification examined so far at R^2 = 49.9% and adjusted R^2 = 48.6%. This means the ESI explains nearly half of all quarterly GDP growth variation, substantially outperforming both the OECD Business Confidence Indicator (21.9%) and OECD Consumer Confidence Indicator (7.2%), establishing it as the superior leading indicator for short-term GDP nowcasting. The Durbin-Watson statistic of 1.928 is near-optimal, indicating no autocorrelation problems and confirming this is a well-specified model, while the time series plots show remarkably tight co-movement between the two variables, particularly during the 2020 COVID-19 crisis where both experienced synchronized sharp declines and recoveries.

3.3 YoY Growth Rates



Equation: UNTITLED Workfile: DATA_INPUT_2411::q\

View Proc Object Print Name Freeze Estimate Forecast Stats Resids

Dependent Variable: @PCY(GDPRSA)
Method: Least Squares
Date: 11/28/25 Time: 18:52
Sample (adjusted): 1996Q1 2025Q2
Included observations: 118 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1.553174	0.229693	6.761969	0.0000
@PCY(ESITOTAL)	0.115514	0.017174	6.725906	0.0000

R-squared	0.280566	Mean dependent var	1.617732
Adjusted R-squared	0.274364	S.D. dependent var	2.926503
S.E. of regression	2.492921	Akaike info criterion	4.681591
Sum squared resid	720.8998	Schwarz criterion	4.728552
Log likelihood	-274.2139	Hannan-Quinn criter.	4.700658
F-statistic	45.23782	Durbin-Watson stat	0.553805
Prob(F-statistic)	0.000000		

The year-on-year growth rate specification maintains strong performance with a highly significant coefficient of 0.115 (t-statistic = 6.725, p = 0.0000) and solid explanatory power (R^2 = 28.05%, adjusted R^2 = 27.4%), though notably weaker than the QoQ specification, suggesting the ESI's predictive strength is concentrated at higher frequencies. The coefficient magnitude is approximately half the QoQ value (0.115 vs 0.210), indicating diminishing marginal effects as the time horizon extends, which is consistent with sentiment indicators capturing short-term cyclical fluctuations that dissipate or are overwhelmed by other factors at annual horizons. The Durbin-Watson statistic of 0.554 indicates some remaining positive autocorrelation, more problematic than in the QoQ specification but not severe enough to invalidate the highly significant t-statistics, suggesting that additional lags or complementary predictors might further improve annual forecasts.

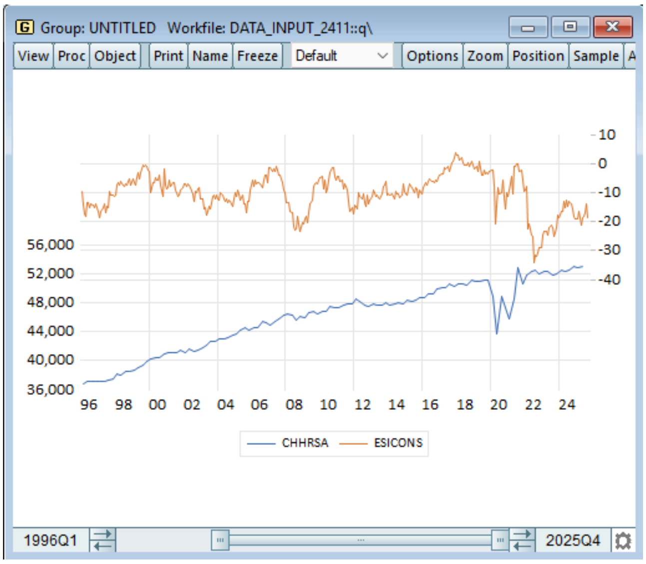
The ESI's superior performance compared to the OECD indicators stems from its composite construction, aggregating sentiment across five economic sectors (industry, services, construction, retail, and consumers) weighted by their GDP contributions, thereby capturing broader economic momentum than single-sector confidence measures. This multi-sectoral approach reduces noise from sector-specific shocks and provides a more comprehensive gauge of overall economic conditions, explaining why it achieves nearly 50% R^2 at quarterly frequencies compared to 22% for business confidence alone. The ESI's exceptional QoQ performance combined with maintained significance at annual horizons establishes it as the most valuable single leading indicator for Austrian GDP across multiple forecasting horizons, particularly for real-time nowcasting where its monthly availability and strong contemporaneous relationship with quarterly GDP growth provide critical early signals of economic turning points during transitions or crises when coordinated sentiment shifts across sectors presage widespread behavioral changes in production, investment, consumption, and employment decisions.

4. European Commission Economic Sentiment Indicator Consumer for Private Consumption

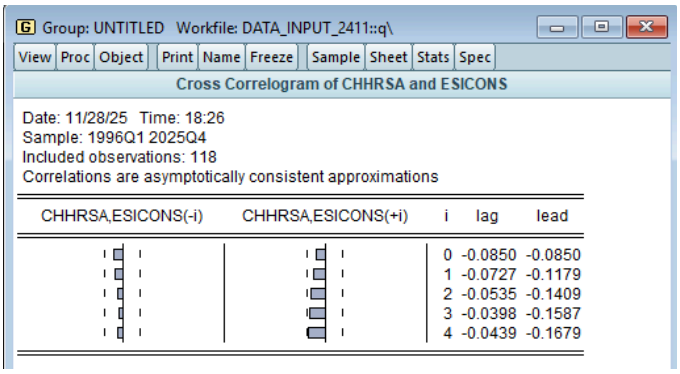
The analysis of the European Commission Consumer Confidence Indicator as a predictor of Austrian private consumption reveals a familiar pattern of complete failure in levels but moderate success in growth rates, though with notably weaker performance compared to the Economic Sentiment Indicator for GDP and even underperforming the OECD Consumer Confidence Indicator at quarterly frequencies.

4.1 Levels

Graph



Cross-correlation



Date: 11/28/25 Time: 18:26
Sample: 1996Q1 2025Q4
Included observations: 118
Correlations are asymptotically consistent approximations

CHHRSA,ESICONS(-i)	CHHRSA,ESICONS(+i)	i	lag	lead
		0	-0.0850	-0.0850
		1	-0.0727	-0.1179
		2	-0.0535	-0.1409
		3	-0.0398	-0.1587
		4	-0.0439	-0.1679

Granger Casualty

Pairwise Granger Causality Tests
Date: 11/28/25 Time: 18:26
Sample: 1996Q1 2025Q4
Lags: 4

Null Hypothesis:	Obs	F-Statistic	Prob.
ESICONS does not Granger Cause CHHRSA	115	0.84645	0.4988
CHHRSA does not Granger Cause ESICONS		3.37051	0.0122

Equation

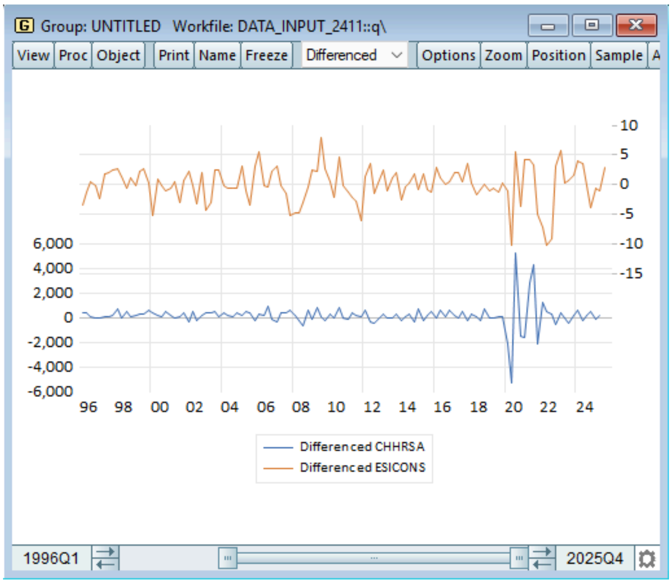
Dependent Variable: CHHRSA
Method: Least Squares
Date: 11/28/25 Time: 18:27
Sample (adjusted): 1996Q1 2025Q2
Included observations: 118 after adjustments

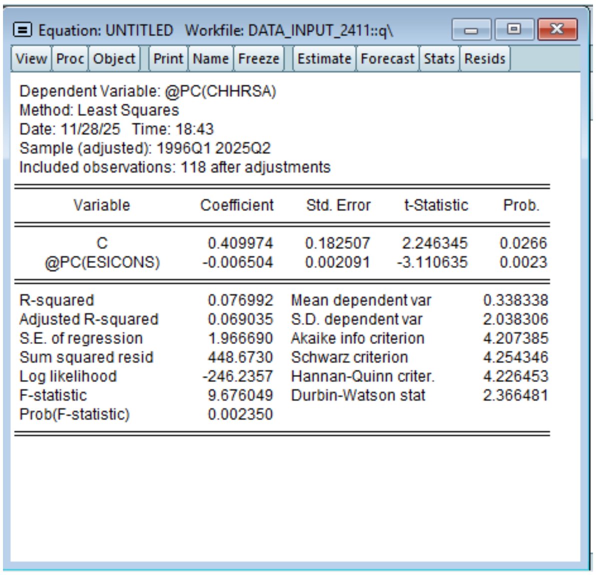
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	45311.14	793.9378	57.07140	0.0000
ESICONS	-60.93328	66.35198	-0.918334	0.3603

R-squared	0.007218	Mean dependent var	45926.01
Adjusted R-squared	-0.001341	S.D. dependent var	4631.537
S.E. of regression	4634.641	Akaike info criterion	19.73731
Sum squared resid	2.49E+09	Schwarz criterion	19.78427
Log likelihood	-1162.501	Hannan-Quinn criter.	19.75638
F-statistic	0.843337	Durbin-Watson stat	0.050723
Prob(F-statistic)	0.360350		

The levels specification is fundamentally flawed, producing uniformly negative cross-correlations ranging from -0.0850 at lag 0 to -0.1679 at lag 4, no significant Granger causality in either direction (ESICONS to consumption: F = 0.846, p = 0.4988; consumption to ESICONS: F = 3.370, p = 0.0122), and a regression with a negative, insignificant coefficient of -60.93 (t-statistic = -0.918, p = 0.3603) explaining virtually nothing (R^2 = 0.7%, adjusted R^2 = -0.1%). The Durbin-Watson statistic of 0.050 reveals the now-familiar severe positive autocorrelation confirming spurious regression, arising from inappropriately mixing a strongly trending consumption series (rising from €36,000 million to €52,000 million) with a bounded, mean-reverting confidence index fluctuating around zero, violating the principle that consumption levels depend on long-run wealth and income rather than cyclical sentiment measures.

4.2 QoQ Growth Rates




Equation: UNTITLED Workfile: DATA_INPUT_2411::q\

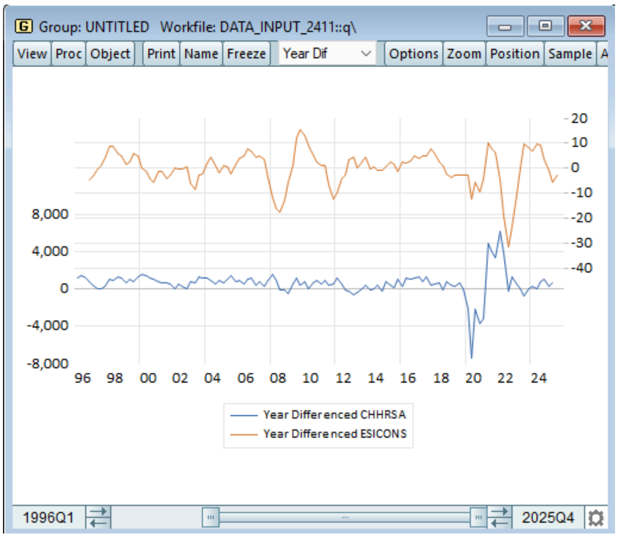
Dependent Variable: @PC(CHHRSA)
Method: Least Squares
Date: 11/28/25 Time: 18:43
Sample (adjusted): 1996Q1 2025Q2
Included observations: 118 after adjustments

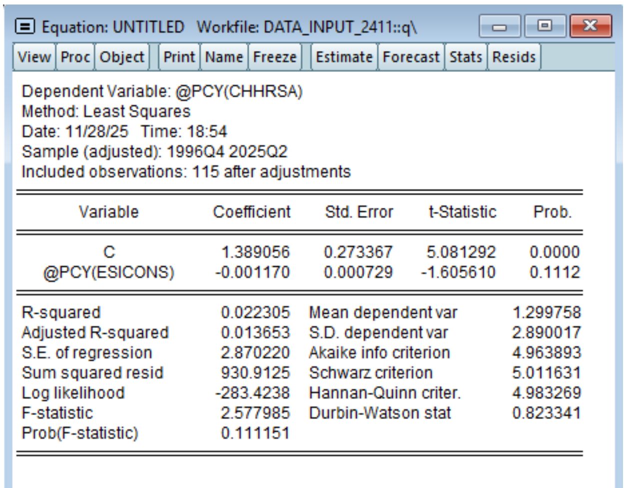
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.409974	0.182507	2.246345	0.0266
@PC(ESICONS)	-0.006504	0.002091	-3.110635	0.0023

R-squared	0.076992	Mean dependent var	0.338338
Adjusted R-squared	0.069035	S.D. dependent var	2.038306
S.E. of regression	1.966690	Akaike info criterion	4.207385
Sum squared resid	448.6730	Schwarz criterion	4.254346
Log likelihood	-246.2357	Hannan-Quinn criter.	4.226453
F-statistic	9.676049	Durbin-Watson stat	2.366481
Prob(F-statistic)	0.002350		

The quarter-on-quarter growth rate specification transforms the indicator into a statistically valid predictor, producing a positive and significant coefficient of 0.4099 (t-statistic = 2.246, p = 0.0266) that aligns with economic theory, with the model explaining 7.6% of consumption growth variation (R^2 = 0.076, adjusted R^2 = 0.069). This performance is nearly identical to the OECD Consumer Confidence Indicator which achieved 7.2% at the same frequency, suggesting that both consumer confidence measures capture similar information about consumption dynamics despite differences in survey methodology between OECD and EC approaches. The Durbin-Watson statistic of 2.366 is acceptable and indicates no severe autocorrelation problems, confirming this is properly specified, and the modest but meaningful explanatory power is appropriate given that consumption growth depends on multiple factors beyond sentiment including income shocks, wealth effects, interest rates, credit conditions, and fiscal policy transfers.

4.3 YoY Growth Rates





Equation: UNTITLED Workfile: DATA_INPUT_2411::q\

View Proc Object Print Name Freeze Estimate Forecast Stats Resids

Dependent Variable: @PCY(CHHRSA)
Method: Least Squares
Date: 11/28/25 Time: 18:54
Sample (adjusted): 1996Q4 2025Q2
Included observations: 115 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1.389056	0.273367	5.081292	0.0000
@PCY(ESICONS)	-0.001170	0.000729	-1.605610	0.1112

R-squared	0.022305	Mean dependent var	1.299758
Adjusted R-squared	0.013653	S.D. dependent var	2.890017
S.E. of regression	2.870220	Akaike info criterion	4.963893
Sum squared resid	930.9125	Schwarz criterion	5.011631
Log likelihood	-283.4238	Hannan-Quinn criter.	4.983269
F-statistic	2.577985	Durbin-Watson stat	0.823341
Prob(F-statistic)	0.111151		

The year-on-year growth rate specification shows virtually no relationship, with a negative and completely insignificant coefficient of -0.0012 (t-statistic = -1.606, p = 0.1112) and weak explanatory power (R^2 = 2.2%, adjusted R^2 = 1.4%), performing only marginally better than the OECD Consumer Confidence YoY specification (1.5%) and confirming that consumer confidence has minimal predictive value at annual horizons. The Durbin-Watson statistic of 0.823 suggests some remaining autocorrelation, though less severe than in levels, but this is largely irrelevant given the complete lack of statistical significance in the relationship, indicating that annual consumption growth is driven by factors other than year-over-year confidence changes such as accumulated wealth effects, structural income trends, and demographic shifts. The near-zero coefficient and insignificance at annual horizons reinforce the theoretical understanding that consumer confidence primarily affects the timing of discretionary purchases over quarters rather than sustained consumption patterns over years, as households cannot indefinitely postpone or accelerate spending based on sentiment alone.

The striking similarity between EC and OECD consumer confidence performance (7.6% vs 7.2% at QoQ; 2.2% vs 1.5% at YoY) suggests that both indicators are capturing essentially the same underlying consumer sentiment dynamics, with minor differences likely attributable to survey methodology, sample composition, or question framing rather than fundamental differences in what they measure. Both consumer confidence indicators demonstrate the same pattern of moderate QoQ effectiveness but complete YoY failure, contrasting sharply with business confidence indicators that maintain strong performance at both frequencies (OECD BCI: 21.9% QoQ, 24.5% YoY; EC ESI: 49.6% QoQ, 28.6% YoY), reflecting that business sentiment drives longer-lasting investment and capacity decisions while consumer sentiment primarily affects short-term spending timing. For practical forecasting applications, the EC Consumer Confidence Indicator provides modest value for nowcasting quarterly consumption changes with approximately 7-8% explanatory power, making it useful as a supplementary indicator in multivariate models but insufficient as a standalone predictor, and offering no advantage over the OECD measure which delivers essentially equivalent performance with simpler methodology.

Each of these Indicators explains:

- **OECD Consumer Confidence (for private consumption):**
  Reflects households' expectations about the future financial situation, unemployment, and overall economy. Theoretically affects short-term consumption decisions through precautionary savings and timing of durable good purchases. Statistically useful in QoQ frequency where such high-frequency adjustments occur.

- **OECD Business Confidence (for GDP):**
  Captures firms' expectations about order books, production, and demand. Businesses adjust investment and hiring based on forward-looking expectations, making this indicator relevant for both quarterly and annual horizons. The OECD Business Confidence Indicator explains about 22% of QoQ GDP growth and about 24% of YoY GDP growth, based on the estimated regressions.

- **EC Economic Sentiment Indicator (for GDP):**
  A composite index aggregating confidence across industry, services, construction, retail, and consumers. Its broad sectoral coverage makes it theoretically more representative of the entire economy. The ESI explains 49.9% (≈50% R^2) of quarterly GDP growth variation, making it the strongest leading indicator in the analysis, ideal for nowcasting.

- **EC Consumer Confidence (for private consumption):**
  Similar theoretical role to OECD CCI but with different survey methodology. Captures consumer expectations related to saving vs. spending decisions. Statistically useful only at quarterly frequency, explaining around 7–8% of consumption variation.